8 CT vs. DT Feedback: Quick Reference

	Continuous-time	**Discrete-time**
Closed-loop formula	$H(s)/[1 + G(s)H(s)]$	$H(z)/[1 + G(z)H(z)]$
Stability boundary	Imaginary axis ($j\omega$)	Unit circle ($\lvert z\rvert = 1$)
Stable poles	$\mathrm{Re}\{s\} < 0$	$\lvert z\rvert < 1$
Stability interpretation	Left-half plane	Inside unit circle

9 Summary

1. $Q = H/(1 + GH)$. The denominator $1 + GH$ determines the closed-loop poles.
2. Simplify block diagrams with three rules: cascade ($\times$), parallel ($+$), feedback loop ($H/(1 + GH)$).
3. Nyquist plot: plot $GH(j\omega)$ in the complex plane. If the curve encircles $-1/K$, the system is unstable (assuming stable open-loop).
4. Gain margin: how many dB of gain you can add before instability (measured at the $-180°$ phase crossing). Phase margin: how many degrees of phase lag you can add before instability (measured at the 0-dB gain crossing).
5. Instability condition: $GH = -1$ (magnitude = 0 dB, phase = $-180°$, simultaneously).

10 Common Mistakes to Avoid

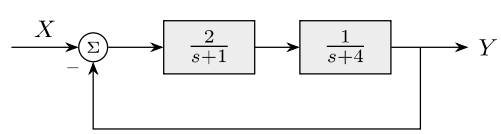
Common Mistake

1. **Sign error at the summer**: negative feedback $\to 1 + GH$; positive feedback $\to 1 - GH$.
2. **Confusing open-loop and closed-loop poles**: the poles of GH are the open-loop poles. The closed-loop poles are the roots of $1 + KGH = 0$, which change with K.
3. **CT vs. DT stability**: left-half plane in CT, inside unit circle in DT.
4. **Reading margins from the wrong curve**: gain margin is read from the *magnitude* plot at the frequency found from the *phase* plot, and vice versa. Do not mix them up.
5. **Forgetting to multiply cascaded blocks**: series $\to$ product, not sum.

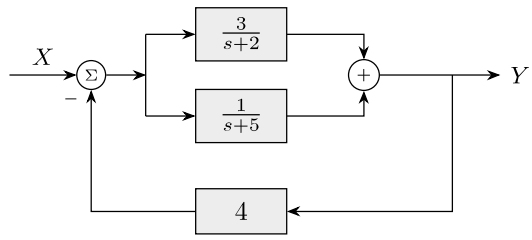
11 Practice Problems

Block Diagram Simplification

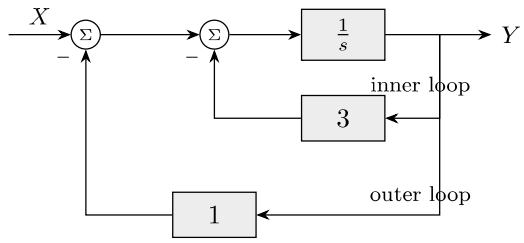
1. **(Cascade + Feedback)** Simplify the following to a single transfer function $Q(s) = Y/X$.



2. **(Parallel + Feedback)** Simplify the following.



3. **(Nested Loops)** Find $Q(s) = Y/X$.



Closed-Loop Poles and Stability

4. A unity-feedback system has $H(s) = K/(s+5)$, $G = 1$.
   (a) Find the closed-loop transfer function and the closed-loop pole.
   (b) For what values of K is the system stable?
   (c) What value of K places the pole at $s = -10$?

5. A feedback system has $H(s) = K/[(s+1)(s+2)]$, $G = 1$.
   (a) Write the characteristic equation (denominator $= 0$).
   (b) For what range of K is the system stable?

6. A DT feedback system has $H(z) = Kz^{-1}/(1 - 0.8z^{-1})$, $G = 1$.
   (a) Find the closed-loop pole as a function of K.
   (b) For what range of $K > 0$ is the system stable?

Gain and Phase Margins

7. From a Bode plot you read:
   - At $\omega_2 = 5$ rad/s (where $|GH| = 0$ dB): $\angle GH = -150°$.
   - At $\omega_1 = 20$ rad/s (where $\angle GH = -180°$): $|GH| = -8$ dB.

   (a) What is the phase margin?
   (b) What is the gain margin (in dB)?
   (c) Is the system stable?
   (d) What is the maximum tolerable additional time delay?

True/False

8. Negative feedback always stabilizes a system.
9. If the phase margin is $60°$, the system is stable.
10. A Nyquist plot is constructed by evaluating $G(j\omega)H(j\omega)$ and plotting it in the complex plane.